GREEEN tallgrass prairie restoration: RLQ analyses

Markus Bauer

2025-08-22

Contents

Preparation 1
Checks . . . 2
Finalize preparation . . . 3

Statistics 3
Descriptive analyses . . . 3
Ordinations . . . 4
RLQ analyses . . . 4
Global test . . . 4
Traits by environment crossed table . . . 4

Plots 5

Save 9

Session info 9

Technichal University of Munich, TUM School of Life Sciences, Chair of Restoration Ecology, Emil-Ramann-Straße 6, 85354 Freising, Germany

*markus1.bauer@tum.de

ORCiD ID: 0000-0001-5372-4174 Google Scholar GitHub: markus1bauer GitHub: TUM-Restoration-Ecology

Preparation

```
library(here)
library(tidyverse)
library(ade4)
```

Packages

```
sites <- read_csv(
  here("data", "processed", "data_processed_sites.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types =
    cols(
    .default = "?",
    seeding_time = "f"
  )) %>%
  arrange(id_plot_year) %>%
```

```r
  filter(
    site == "Lux Arbor",
    richness_type == "seeded_richness",
    treatment_id %in% c("1", "2", "3", "4")
  ) %>%
  mutate(
    treatment = str_c(seeding_time, herbicide, seeded_pool, sep = "_"),
    treatment = factor(treatment)
  ) %>%
  filter(treatment != "unseeded_0_0") %>%
  select(
    id_plot_year, year, treatment, water_cap, cover_non_seeded,
    cover_seeded_grass
    )

traits <- read_csv(
  here("data", "processed", "data_processed_traits.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(.default = "?")
  ) %>%
  filter(seeded == "1", family != "Poaceae") %>%
  select(accepted_name, sla, height, seedmass) %>%
  mutate(seedmass = log(seedmass)) %>%
  drop_na()

species <- read_csv(
  here("data", "processed", "data_processed_species.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(.default = "?")
  ) %>%
  filter(plot_size %in% c("1", "25")) %>%
  mutate(abundance = 1) %>%
  select(id_plot_year, accepted_name, abundance) %>%
  semi_join(sites, by = "id_plot_year") %>%
  semi_join(traits, by = "accepted_name") %>%
  arrange(id_plot_year)
```

Load data

Checks

```r
species %>%
  group_by(id_plot_year) %>%
  count()
```

Seeded species richness

```
## # A tibble: 104 x 2
## # Groups:   id_plot_year [104]
##    id_plot_year     n
##    <chr>        <int>
##  1 Lux_12_2015      9
##  2 Lux_12_2016     10
##  3 Lux_12_2017     10
##  4 Lux_12_2018      9
##  5 Lux_12_2019      8
```

```
##  6 Lux_12_2025      4
##  7 Lux_17_2015      3
##  8 Lux_17_2016      5
##  9 Lux_17_2017      9
## 10 Lux_17_2018      8
## # i 94 more rows
```

```
traits %>% anti_join(species, by = "accepted_name")
```

Check non-occuring seeded species

```
## # A tibble: 5 x 4
##   accepted_name                   sla height seedmass
##   <chr>                         <dbl>  <dbl>    <dbl>
## 1 Dalea purpurea                191.   0.91     -6.07
## 2 Liatris cylindracea           110.   0.564    -5.73
## 3 Lupinus perennis              167.   0.326    -3.84
## 4 Solidago juncea                13.3  0.783    -7.79
## 5 Symphyotrichum oolentangiense  14.3  0.362    -8.68
```

Finalize preparation

```
traits <- traits %>%
  semi_join(species, by = "accepted_name") %>%
  column_to_rownames(var = "accepted_name")

sites <- sites %>%
  semi_join(species, by = "id_plot_year") %>%
  column_to_rownames(var = "id_plot_year")

species <- species %>%
  pivot_wider(
    names_from = "accepted_name", values_from = "abundance", values_fill = 0
    ) %>%
  column_to_rownames(var = "id_plot_year")
```

Statistics

Descriptive analyses

```
sites %>%
  group_by(year) %>%
  count()
```

```
## # A tibble: 6 x 2
## # Groups:   year [6]
##    year     n
##   <dbl> <int>
## 1  2015    14
## 2  2016    18
## 3  2017    18
## 4  2018    18
## 5  2019    18
```

```
## 6  2025    18
```

Ordinations

```r
dudi_l <- dudi.coa(species, scannf = FALSE)
dudi_r <- dudi.hillsmith(
  sites,
  row.w = dudi_l$lw,
  scannf = FALSE
  )
dudi_q <- dudi.pca(
  traits,
  row.w = dudi_l$cw,
  scannf = FALSE
  )
```

RLQ analyses

```r
m1 <- rlq(
  dudiR = dudi_r,
  dudiL = dudi_l,
  dudiQ = dudi_q,
  scannf = FALSE
  )
```

Global test

```r
randtest(m1, nrepet = 999, modeltype = 6)
```

```
## class: krandtest lightkrandtest
## Monte-Carlo tests
## Call: randtest.rlq(xtest = m1, nrepet = 999, modeltype = 6)
##
## Number of tests:   2
##
## Adjustment method for multiple comparisons:   none
## Permutation number:   999
##      Test        Obs    Std.Obs   Alter Pvalue
## 1 Model 2 0.07776746 10.2595009 greater  0.001
## 2 Model 4 0.07776746  0.3117887 greater  0.328
```

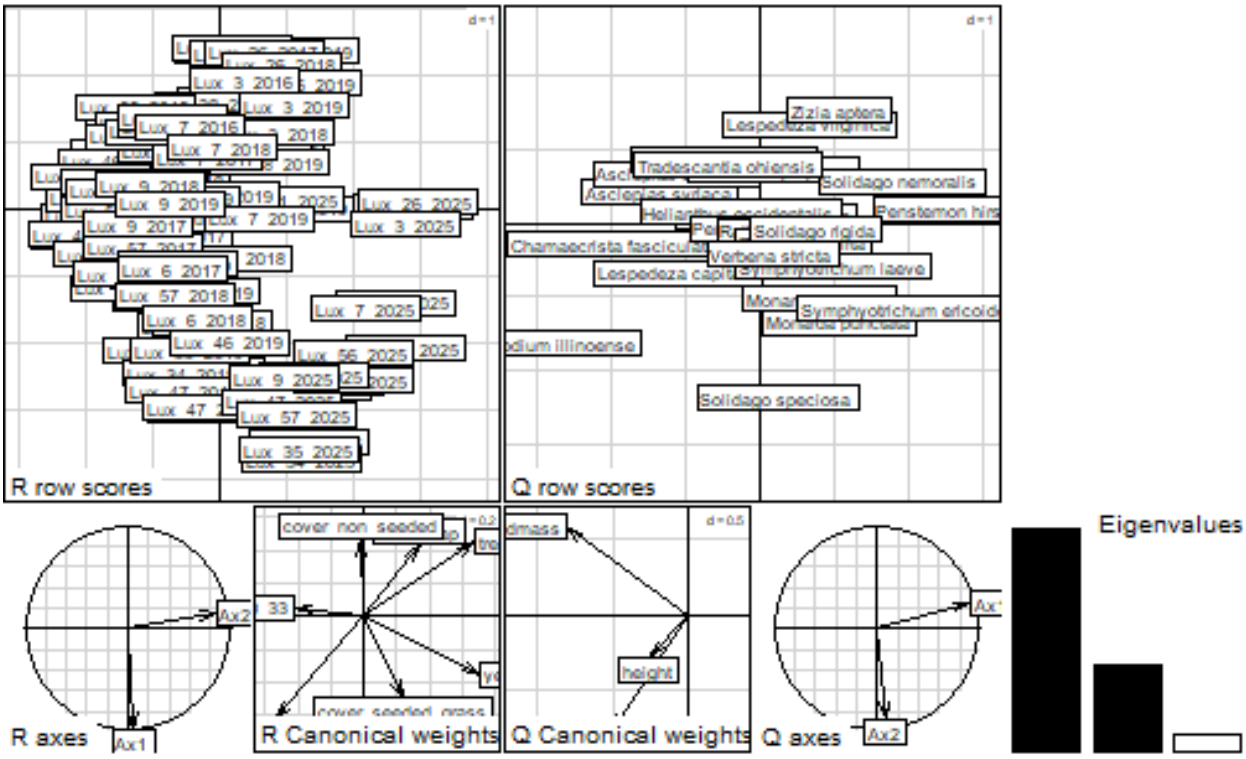
Traits by environment crossed table

```r
m1$tab
```

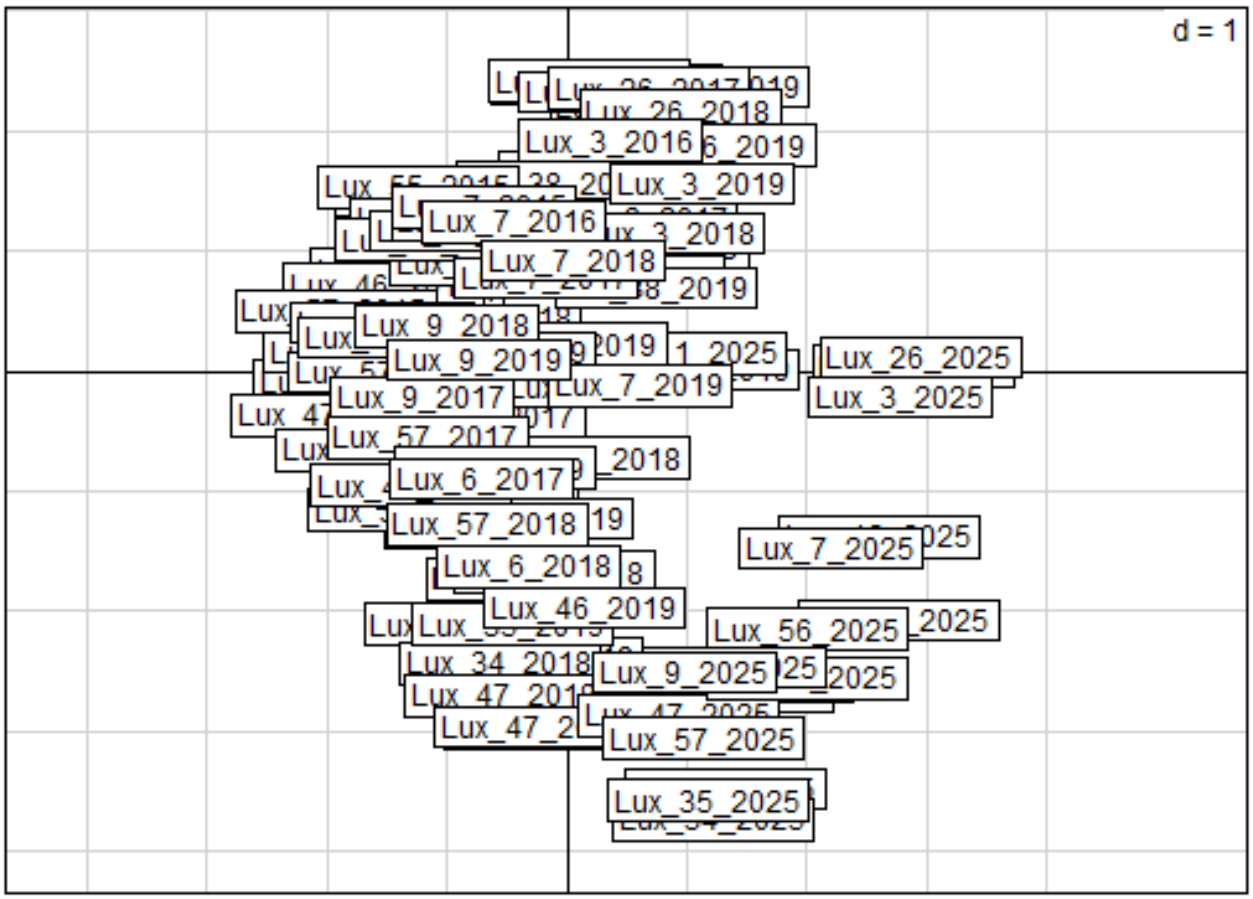
```
##                            sla       height    seedmass
## year               -0.04090369 -0.040676342 -0.16092284
## treat.fall_0_33    -0.14499944 -0.025114547 -0.09172420
## treat.spring_0_33   0.08268829 -0.067648000  0.08891354
## treat.spring_1_33   0.12068355  0.087156858  0.04675457
## water_cap          -0.08327588 -0.050840611 -0.02646510
## cover_non_seeded   -0.04714453 -0.023455511  0.04345643
## cover_seeded_grass  0.02968349 -0.002851539 -0.08573552
```

Plots

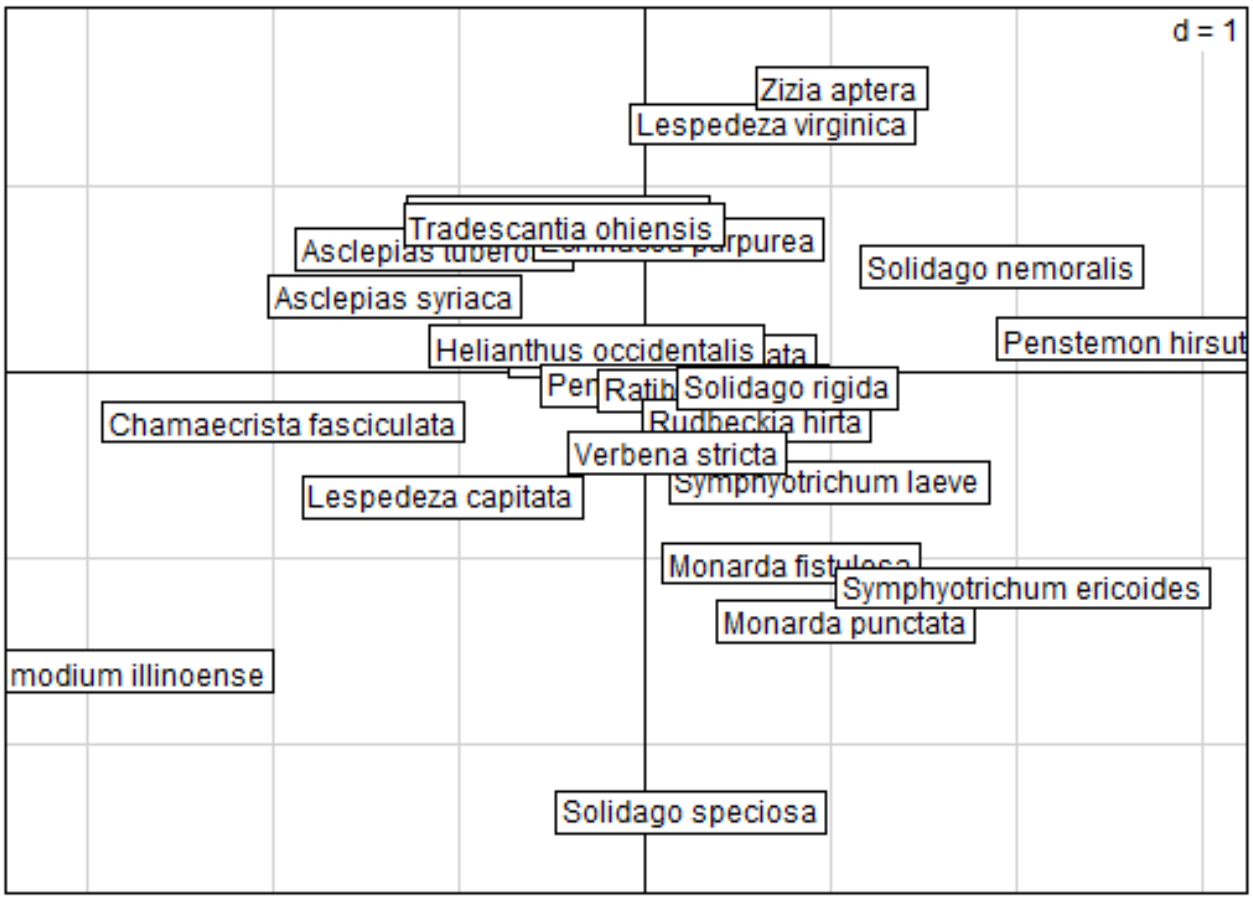
```
plot(m1)
```



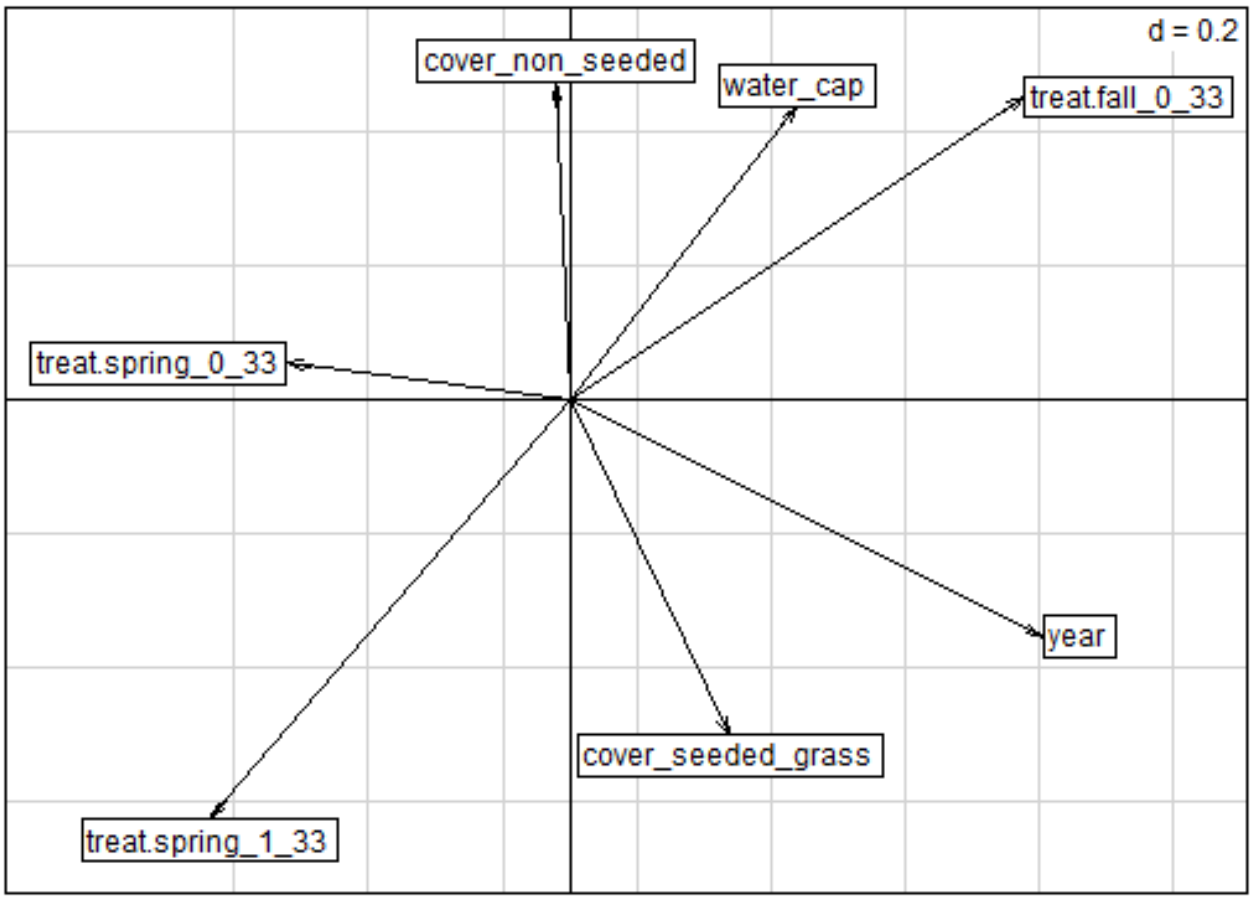

```
s.label(m1$lR)
```


d = 1
Lux_26_2018
Lux_3_2016
Lux_3_2019
Lux_7_2016
Lux_7_2018
Lux_9_2018
Lux_9_2019
Lux_26_2025
Lux_3_2025
Lux_9_2017
Lux_7_2019
Lux_57_2017
Lux_6_2017
Lux_57_2018
Lux_7_2025
Lux_6_2018
Lux_46_2019
Lux_56_2025
Lux_34_2018
Lux_9_2025
Lux_57_2025
Lux_35_2025

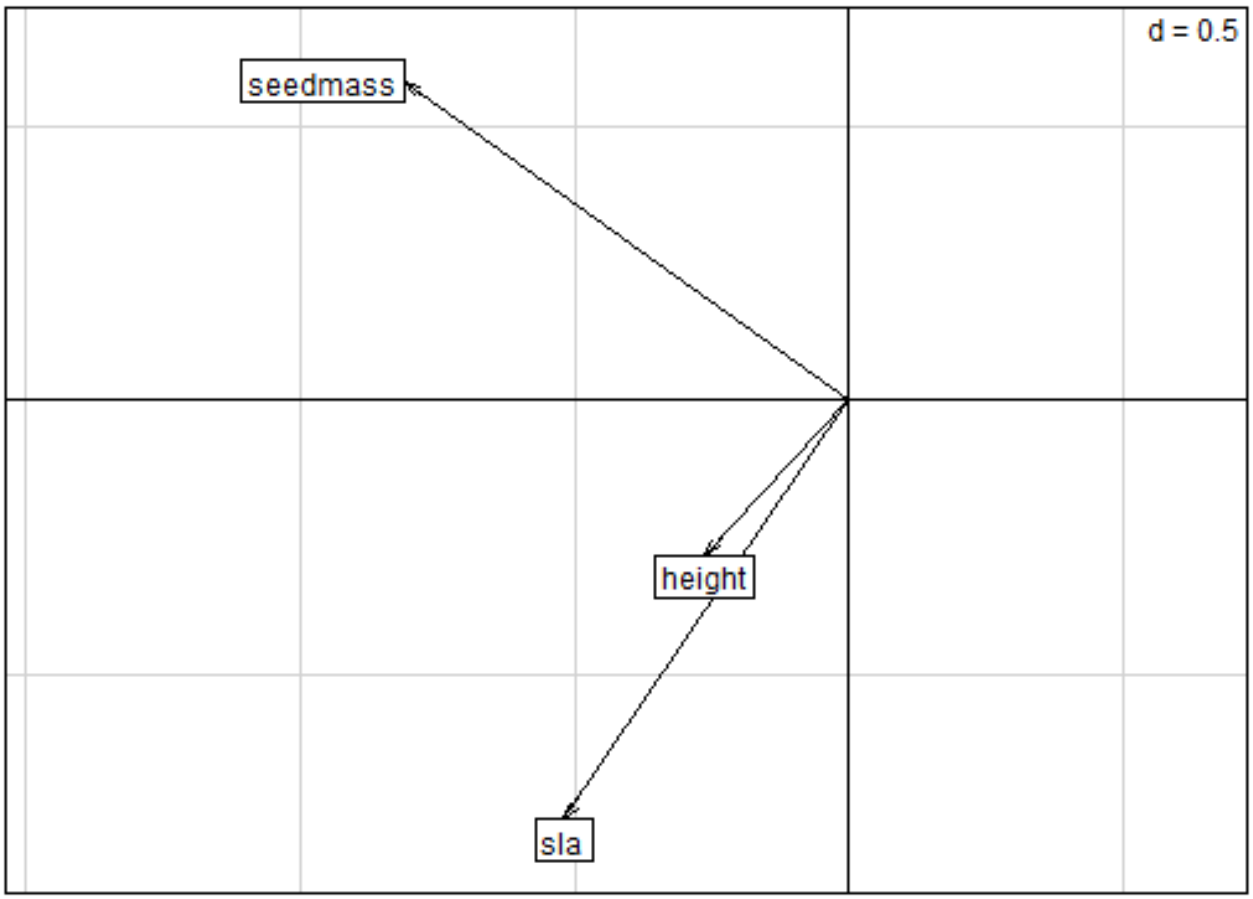
```
s.label(m1$lQ)
```


d = 1
Zizia aptera
Lespedeza virginica
Tradescantia ohiensis
Solidago nemoralis
Asclepias syriaca
Penstemon hirsut
Solidago rigida
Chamaecrista fasciculata
Rudbeckia hirta
Verbena stricta
Lespedeza capitata
Symphyotrichum laeve
Symphyotrichum ericoides
Monarda punctata
modium illinoense
Solidago speciosa

```
s.arrow(m1$l1)
```


d = 0.2
cover_non_seeded
water_cap
treat.fall_0_33
treat.spring_0_33
year
cover_seeded_grass
treat.spring_1_33

```
s.arrow(m1$c1)
```



Save

Session info

```
## R version 4.5.0 (2025-04-11 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26100)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] ade4_1.7-23     lubridate_1.9.4 forcats_1.0.0   stringr_1.5.1
```

```
##  [5] dplyr_1.1.4     purrr_1.1.0     readr_2.1.5     tidyr_1.3.1
##  [9] tibble_3.3.0    ggplot2_3.5.2   tidyverse_2.0.0 here_1.0.1
## [13] tinytex_0.57
##
## loaded via a namespace (and not attached):
##  [1] bit_4.6.0          gtable_0.3.6       crayon_1.5.3       compiler_4.5.0
##  [5] Rcpp_1.1.0         tidyselect_1.2.1   parallel_4.5.0     scales_1.4.0
##  [9] yaml_2.3.10        fastmap_1.2.0      R6_2.6.1           generics_0.1.4
## [13] knitr_1.50         MASS_7.3-65        rprojroot_2.0.4    pillar_1.11.0
## [17] RColorBrewer_1.1-3 tzdb_0.5.0         rlang_1.1.6        utf8_1.2.6
## [21] stringi_1.8.7      xfun_0.52          bit64_4.6.0-1      timechange_0.3.0
## [25] cli_3.6.5          withr_3.0.2        magrittr_2.0.3     digest_0.6.37
## [29] grid_4.5.0         vroom_1.6.5        rstudioapi_0.17.1  hms_1.1.3
## [33] lifecycle_1.0.4    vctrs_0.6.5        evaluate_1.0.4     glue_1.8.0
## [37] farver_2.1.2       rmarkdown_2.29     tools_4.5.0        pkgconfig_2.0.3
## [41] htmltools_0.5.8.1
```